



Certificate of Measured Result

CounterFact 1,000-edit comparison on GPT-J-6B · Ed25519 signed · independently verifiable

GIC-COUNTERFACT-GPTJ-2026-001 · issued 2026-09-09 · signed 2026-09-09T13:10:14Z · self_sha256 7a41ffa4ad9a6caa...

MODEL IDENTITY

Field	Value
Base model	EleutherAI/gpt-j-6B (base)
Base weights SHA256	e90ae6645c7a681df43259a9ccc09ea7c5b0a213105e1f7e9275a87ee9bb7ed4
Base weights note	EleutherAI/gpt-j-6B, unmodified, scored as bf16 (fp32 checkpoint cast once; no rounding precedes any edit)
Certified weights (Probe56)	e90ae6645c7a681df43259a9ccc09ea7c5b0a213105e1f7e9275a87ee9bb7ed4 unmodified = base hash — no edit, identical hash
Document	CounterFact 1,000-edit comparison on GPT-J-6B — ROME / MEMIT / AlphaEdit (EasyEdit, shipped defaults) vs Probe56 runtime hooks
Edits / controls / held-out	1,000 edits · 500 controls · 5,209 held-out questions
Controls floor	1 of 500 controls was not rank-1 on the unmodified base; collateral is counted above that floor (499 known controls)
Harness	rank-1 of the gold token over the full vocabulary; bf16 backbone, fp32 lm_head, left padding with padding-aware position ids, left-truncation to 2,048 tokens applied identically in every pass, batch 24, fixed order; identical script for every run (harness/cf_harness_gptj.py)
Harness SHA256	174717075ab79272de76c4fcc9d1db7f962544b96aced0aba9c87a80a838f49b
Truncation	left-truncation to 2,048 tokens, applied to every pass of every method, base included; 388 held-out prompts exceed the limit (longest 2,874 tokens)

CERTIFIED RESULT — ALL METHODS, ONE HARNESS

Method	Edits at rank 1	Success	Nearby facts broken	Held-out at rank 1	Δ held-out
Probe56	985 / 1,000	98.5%	0 / 499	930 / 5,209	+0.00 pp
ROME (1,000 edits)	0 / 1,000	0.0%	499 / 499	0 / 5,209	-17.85 pp
ROME (dose point 1)	6 / 1,000	0.6%	251 / 499	904 / 5,209	-0.50 pp
ROME (dose point 10)	3 / 1,000	0.3%	499 / 499	85 / 5,209	-16.22 pp
MEMIT	990 / 1,000	99.0%	110 / 499	937 / 5,209	+0.13 pp
AlphaEdit	987 / 1,000	98.7%	143 / 499	922 / 5,209	-0.15 pp

Probe56: 985 of 1,000 edits at rank 1, 0 of 499 nearby facts broken, 930 of 5,209 held-out questions still at rank 1 — identical to the unmodified base (930), +0.00 pp. The control result is a census over all 500 controls with every hook active, not a sample; no confidence interval is quoted because nothing was sampled. Weights are byte-identical to the frozen base.

HELD-OUT EVIDENCE BY BENCHMARK

Benchmark	Questions	Base rank-1	Probe56 Δ	ROME Δ
arc_challenge	391	18.67%	+0.00 pp	-18.67 pp
gpqa_diamond	98	30.61%	+0.00 pp	-30.61 pp
gsm8k	440	27.95%	+0.00 pp	-27.95 pp
mmlu_pro	4,010	15.81%	+0.00 pp	-15.81 pp
truthfulqa_mc2	270	25.93%	+0.00 pp	-25.93 pp
TOTAL	5,209	17.85%	+0.00 pp	-17.85 pp



Cryptographic Provenance

GIC-COUNTERFACT-GPTJ-2026-001 · Ed25519 RPNY2byY9Ne0LAMD...

HASH CHAIN

The chain binds the frozen base weights, each method's edited weights, and every evidence artifact to this certificate by hash. It proves the Probe56 result was measured on unmodified weights, without describing the method.

```
BASE WEIGHTS (EleutherAI/gpt-j-6B (base), frozen)
sha256: e90ae6645c7a681df43259a9ccc09ea7c5b0a213105e1f7e9275a87ee9bb7ed4
└── weights unmodified: no edit, identical hash ───>
CERTIFIED WEIGHTS (Probe56)
sha256: e90ae6645c7a681df43259a9ccc09ea7c5b0a213105e1f7e9275a87ee9bb7ed4

EDITED WEIGHTS (EasyEdit commit 14cea8245f06...) - each 12,101,796,792 bytes
ROME      sha256: d39a5c469eeb9b63033b4d05ac439ba5d8a0d0a416674d5d956f5242273c1a32
MEMIT     sha256: 91a2d65aa0f7e9bf24ffe2ba24533dd417c7daa968dbc81a2d9e3c0a4f3cf8f6
AlphaEdit sha256: 7449cc9fbd79bf87c59288bdecfa2d5bbf8cdf6d5f1cfeccdf681ba4642975ed

MERKLE ROOT (sha256 over the sorted lines 'path:sha256\n' of file_hashes_sha256; 29 files)
sha256: 91785e98e1ec6c8036ff22107fc4598adf54d02ef3271a1968c18e065b5121b1

CERTIFICATE (this document's source JSON)
self_sha256 (canonical bytes): 7a41ffa4ad9a6caa0f87013ae3c88e2b8a3cfecbdd4c2e7fe6c29057207bb58
file sha256: 7273c527c304cbc809dc3deac31d075594917f8ce2eadd3c4964fc08397f634c
```

WEIGHTS REGISTRY

Run	Artifact	SHA256
Base	combined over base files	e90ae6645c7a681df43259a9ccc09ea7c5b0a213105e1f7e9275a87ee9bb7ed4
Probe56	combined over base files	e90ae6645c7a681df43259a9ccc09ea7c5b0a213105e1f7e9275a87ee9bb7ed4
ROME	model.safetensors	d39a5c469eeb9b63033b4d05ac439ba5d8a0d0a416674d5d956f5242273c1a32
ROME1	model.safetensors	5c16bc23e22281c9c38f7306d383d709f6192f91b6cc8f97e2be06a3e8a2c900
ROME10	model.safetensors	34f97ee4cafcd995825a35ee2ba282359c8560baa334dc9ab3be0e281ab1dc5
MEMIT	model.safetensors	91a2d65aa0f7e9bf24ffe2ba24533dd417c7daa968dbc81a2d9e3c0a4f3cf8f6
AlphaEdit	model.safetensors	7449cc9fbd79bf87c59288bdecfa2d5bbf8cdf6d5f1cfeccdf681ba4642975ed



Signature and Verification

GIC-COUNTERFACT-GPTJ-2026-001 · Ed25519 RPNY2byY9Ne0LAMD...

ED25519 SIGNATURE

SIGNATURE (base64)

RPNY2byY9Ne0LAMDzViMbGrJXysgkpmo6H07rMI/ZT8rz0r8utv0XPvZkEuTn1AfUseykHvJ5nZtoKkEiPC3Dg==

PUBLIC KEY (Founder)

FtrWshUc/9rg5Cz+ARi5DP/yyqFhWMJLmx0VHKc3wpk=

MERKLE ROOT

91785e98e1ec6c8036ff22107fc4598adf54d02ef3271a1968c18e065b5121b1

SIGNED MESSAGE

canonical JSON of this file with the 'signature' and 'self_sha256' fields removed: json.dumps(obj, sort_keys=True, separators=(',', ':'), ensure_ascii=True) encoded UTF-8. ensure_ascii=True (non-ASCII characters escaped as \uXXXX) is part of the rule; the document text contains non-ASCII characters, so serialising without it produces different bytes and the signature will not verify.

VERIFY

1. Download the proof package.
 2. Compute SHA256 of every file and match SHA256SUMS.txt and file_hashes_sha256.
 3. Recount the numbers from runs/*/... (rank == 1 per row) and match 'numbers'.
 4. Recompute merkle_root_sha256 per merkle_definition.
 5. Verify the Ed25519 signature against the public key FtrWshUc/9rg5Cz+ARi5DP/yyqFhWMJLmx0VHKc3wpk=. The signed bytes are this JSON with the 'signature' and 'self_sha256' fields removed, serialised as canonical JSON: sorted keys, separators=(',', ':'), ensure_ascii=True (non-ASCII characters escaped as \uXXXX), encoded UTF-8 — in Python, json.dumps(obj, sort_keys=True, separators=(',', ':'), ensure_ascii=True).encode('utf-8'). ensure_ascii=True is part of the rule: this document contains non-ASCII characters, so serialising without it produces different bytes and the signature will not verify. 'self_sha256' is the sha256 of those same bytes.
 6. Or run the script that does all of the above: python verify_certificate.py CounterFact-certificate-SIGNED.json (needs only the 'cryptography' package).
- python verify_certificate.py, shipped in the same proof directory, performs steps 2–5 and prints one verdict line. Expected output on an unmodified package: signature VALID, self_sha256 matches, 29 of 29 files verified, merkle_root matches, RESULT OK.

Growing Intelligence · yonathan@growing-intelligence.com · growing-intelligence.com

This document records a named reproducible procedure and its measured results. It is not an accreditation and invokes no external standards body. The Probe56 procedure is protected by provisional patents filed with the USPTO and is not described here.



Evidence Manifest

29 files · merkle root 91785e98e1ec6c80...

FILE HASHES (SHA256)

Path	SHA256
harness/cf_harness_gptj.py	174717075ab79272de76c4fcc9d1db7f962544b96aced0aba9c87a80a838f49b
packages/AlphaEdit/MANIFEST.json	b0c79cf86086bfecdd3da9942143959979c5ca1b79b71db853bc8c9e2331c2925
packages/MEMIT/MANIFEST.json	1fc9186d7cb834fe854d2e848a03d48bb56f6bd0a1a07cd5600a761e31e32aeb
packages/ROME/MANIFEST.json	8feedfffc3691d427bfabd6160dca4c789c69c602592bf41bffc27158883a41d
packages/ROME1/MANIFEST.json	88aed9d2ff7bfb452cedeeab9512fd8d78b59216c18219dd2c28f774fdbf31a6
packages/ROME10/MANIFEST.json	d9634fa2001645d15e14ed3dfd296671b50ffa2e449ebb38ca320b471573102f
probe56/census_summary.json	f1d000adcbc8f1dda23ae5009fc30cc3b21a2333995ab04d29d1a1070f0c8f76
runs/AlphaEdit/controls_results.jsonl	f9cdbfa9b80f47bbfd14cd047b16411f1419510967b7c727305962b366b71a37
runs/AlphaEdit/edits_results.jsonl	b205dc662c00dacac107b0cf2e2f7d6b4085e344268b78363a0ccba43da0faec
runs/AlphaEdit/heldout_results.jsonl	efc9c31d0eda391c7f6ec34b36d7eb1b16e34fb4da0d214eb933101ee4a892aa
runs/Base/controls_results.jsonl	2b9131ac917ff9ea97cf360d037326a241f9a8619324fab4f595197ec8199c0e
runs/Base/edits_results.jsonl	ad98274f1dc7a271bbc3ff3b0df566cfe6499f5a9ef263e27a83f8fd4fa9f04d
runs/Base/heldout_results.jsonl	5d20913cc906d1f90feaf1eb5f97bacffa61dc530ad223c626d3cb5bd960f919
runs/MEMIT/controls_results.jsonl	1119fc360da3908fca4c914ffcb1ebfa7836a3cc1f40784fce13512c073420aa5
runs/MEMIT/edits_results.jsonl	d4cedafee7ff0c2ad89a2f0682757993a35bba894f248de1c912b83f1305bbf8
runs/MEMIT/heldout_results.jsonl	44e0abd48b9b66588f45f1f699d541073e29f245c7dc4c8add6fb74cd0949ba9
runs/Probe56/controls_results.jsonl	2b9131ac917ff9ea97cf360d037326a241f9a8619324fab4f595197ec8199c0e
runs/Probe56/edits_results.jsonl	8a6599334ae3a2916dd21df45ad11fd44ee975860f46b3dcc406557672bc4a95
runs/Probe56/heldout_results.jsonl	5d20913cc906d1f90feaf1eb5f97bacffa61dc530ad223c626d3cb5bd960f919
runs/ROME/controls_results.jsonl	f6ab7bba57b2c4b710f347672193dfba538beacadaf1e6782033e5ea969f657f
runs/ROME/edits_results.jsonl	5c41a719c48bd402b34c7c19a69778cbdc0f91ffb48e8dffbf7ff6aafe1d6b180
runs/ROME/heldout_results.jsonl	aad8473489fb0881e36e2e26113c9a0b62a875bf4525e95482fe2468dc5b5746
runs/ROME1/controls_results.jsonl	9b4a0383faf4b64a08a92ebc79c78074efff99d3882170dfd83424a0cdfad57b
runs/ROME1/edits_results.jsonl	3d92b29a242ebd7aac3adf0b69e753acbd8f29da803731f0e13d5ae6695191f1
runs/ROME1/heldout_results.jsonl	790d198eed144a32210dba2ef869ced9757c65e51fb2cf0af62d621f34702e78
runs/ROME10/controls_results.jsonl	86007d09edfc597c0b771a6cd64ea1f7bb8ff4f2930bcd028d3775eff0318eb4
runs/ROME10/edits_results.jsonl	73a6c172c72394af10de4315c84bd79a901974f112ad1aafd4b921bc77ee47af
runs/ROME10/heldout_results.jsonl	238020313c65521b078b504f4a27ee652106cff82f95b6f763c6ba9083bd5884
zero_edit_control/DTYPE_CONTROL.json	fb094e9ee5998b5e371ad252d02b1eeec68c966c0563a8a0a2dae13760f918cd

The Merkle root is sha256 over the sorted lines 'path:sha256\n' of file_hashes_sha256. Recomputing it over this table must reproduce 91785e98e1ec6c8036ff22107fc4598adf54d02ef3271a1968c18e065b5121b1.